

Detection & Classification of Spoofing Attacks on GNSS Signals Using AI-Based Models on NavIC System.

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1. Abstract

Global Navigation Satellite Systems (GNSS) have become an indispensable part of our contemporary infrastructure for navigation and timing, being employed in several mission critical applications ranging from unmanned aerial vehicles (UAVs), autonomous transportation systems to aviation, maritime navigation and communication networks. Nevertheless, even when there exists such a widespread use, GNSS is still afflicted by weak authenticity of transmitted signals (spoofing); that is an attacker could send counterfeit signals, which can deceive the receiver involved in calculating wrong position or velocity or timing information. These attacks represents a great risk especially in safety critical contexts where the non detection of even an only spoofing is enough to cause catastrophic events.

In this paper, we propose an integrated artificial intelligence based framework for GNSS spoofing attack detection on real signal data. Three sophisticate machine learning models — Deep Neural Networks (DNN), Random Forest (RF) and Extreme Gradient Boosting (XGBoost) — were constructed, trained and tested to see how well they could distinguish genuine and spoofed signals. The achieved cross-sections are based on realistic data sets containing environmental noise, multipath effects and interference, which in contrast to simulated data better demonstrate a truly practical disposition.

The proposed structure is based on sophisticated feature engineering obtained from core GNSS signal parameters such as carrier-to-noise density ratio (C/N_0), Doppler shift, pseudorange error and temporal stability qualities. Experimental results show that three models are all over the 93% accuracy, but XGBoost is superior to others, and can achieve 95.5% of the accuracy as well as reduce a lot false negative rates (the biggest danger for security in LGKS).

In addition, it provides a potential action plan for the real-time implementation of XGBoost model into UAV navigation equipment so as to monitor GNSS signals in real time and even take necessary actions automatically when attacks are identified, like navigational method switchover to inertial navigation or launching protection process. Manipulation shows that data-driven AI-based spoofing detection systems, trained on actual-data and reinforced with advanced feature manipulation, are a practical and scalable means for augmenting the security of GNSS applications in future autonomous systems.

2.Introduction

Global Navigation Satellite Systems (GNSS) have become critical to the functioning of our modern technology-based societies through Positioning, Navigation and Timing (PNT). These services drive civilian and military applications ranging from smartphones navigation and fleet management to aviation essential systems, maritime navigation, as well as controlling autonomous unmanned platforms. Because critical and safety-relevant applications are increasingly based on GNSS there is a growing need to achieve integrity and reliability of satellite-based navigation.

And finally, GNSS signals, while being a great technology to have offer very low-powered signals once they reach ground level and are susceptible. Spoofing attacks are especially harmful, since such attacks do not typically bring down the service, but instead lead the receiver to accept incorrect yet believable navigation information. Thus, the attacked entity may function properly relying on invalid positional or temporal information and therefore, such kinds of attacks are specifically challenging to discover.

Spoofing detection via Signal Processing: The diagnostic techniques for detecting GNSS spoofing have remained largely unchanged over the years and have included a blended use of signal processing (e.g., signal power, angle-of-arrival, cryptographic authentication, and multi-sensor consistency checks). Although successful in some cases, they typically involve costly hardware, multiple antennas, or knowledge of attack profile. What is more, their accuracy may weaken in complex environments with multipath propagation, dynamic motion, or advanced attacks that closely replicate the behaviors of legitimate signals.

The rapid development of artificial intelligence and machine learning capabilities hold new potential to develop GNSS spoofing mitigation approaches that are able to adapt and learn from patterns and experiences. In general, data-driven models can learn intricate and nonlinear relationships between signal features that may not be easily captured by rule-based methods. Machine learning models accomplish this by learning earlier signal patterns and identifying the minutest of variations to statistical characteristics caused by spoofing even when counterfeit signals have almost similar characteristics to authentic signals.

In this light this paper aims to explore the use of complex machine learning models (DNN, RF and XGBoost) in GNSS spoof detection from a practical standpoint, using real world data. Special attention is paid to reducing false-negative mistakes taking

into consideration their negative and immediate impact on navigation safety. Based on systematic comparative analysis and real-time

3. Aim and Objectives

This research is oriented towards building, developing and testing as an artificial intelligence-based framework of the essential points necessary for the operational reliability in applied real-life experiences and in real time regarding the performance of spoofing attacks on Global Navigation Satellite System (GNSS) applications. This work aims to overcome the classic spoofing detection approach limitation by using advanced learning machine models, which can learn more complex signal behavior in nominal and spoofed environments.

The motivation behind this work is the recent growth of GNSS dependency in safety-critical applications such as unmanned aerial vehicles, autonomous navigation systems and timing-critical infrastructures, where even a single undetected spoofing attack can lead to catastrophic operational and security consequences.

Therefore, the goal is not just to maximize the classification accuracy, but also to minimize false-negative errors for reliable and timely detection of spoofing attacks across varying environmental conditions.

In order to fulfil this aim, the following specific objectives are set:

Goal 1: Building a Non-ideal GNSS Spoofing Detection Dataset

The last of these three is more concerned with authentic and spoofer GNSS signal data usage and processing. RAW data omit environmental noise, multipath, receiver imperfections, and time-varying signals, which makes the proposed detection framework characterized close to real operation condition.

Goal 2: Engineering features and characterizing the signal

Its goal is to obtaining and creating informative features from unique view GNSS signals measurements. Essential factors are the C/N_0 (carrier-to-noise density ratio), Doppler shift, pseudorange error, and position changes. A number of extra features including motion statistics, signal stability indicators and local variation statistics are computed to capture fine spoofing patterns unable to be learned by the models. Design and Development of Multiple AI Models.

Goal.3:

To achieve this goal, several machine learning models can be designed and implemented, specifically Deep Neural Networks (DNN), -to investigate the merits and demerits of each method within GNSS spoofing detection and to perform a comparative analysis across learning architectures.

Goal. 4: Comparison + Error Analysis

The goal of this task is the systematic comparison of model performance with standard metrics such as accuracy, recall, precision, and the F1-score needed for confusion matrices. The analysis of false-negative errors is focused upon because any such errors directly affect navigation system safety.

Goal 5: Evaluation for Generalization and Robustness

This goal assesses the trained models' performance to generalize over new data and over different signal conditions (signal power, Doppler error as well as noise), entailing the need for stable behavior in a wide range of operational environments.

Goal 6 :Exploration of real-time deploy ability and end-to-end system integration Photo by Jesse Gardner on Unsplashd

The last objective is to assess the potential of incorporating the most successful model into real-time GNSS-receiver based

UAV platforms and other navigation systems. This entails assessing response latency, determining the amount of computational resources needed, and suggesting automated mitigation techniques to allow for quick action in the event that a spoofing attack is discovered.

Together, these goals seek to create a thorough, scalable, and intelligent framework for GNSS spoofing detection that connects theoretical studies with real-world, operational applications in the field of satellite navigation security.

4.Methodology

This chapter outlines the full methodological approach used in this research to detect spoofing attacks on Global Navigation Satellite System (GNSS) signals through the use of artificial intelligence and machine learning. The methodology is structured to ensure that the results can be replicated, that different models can be fairly compared, and that the approach can be applied effectively in real-world settings. It covers all the key stages, including data collection, preprocessing, feature engineering, model development, training strategies, and methods for evaluating performance.

4.1 Data Collection and Dataset Description

The research employs real-time GNSS signal datasets that are comprised of real as well as spoofed GNSS signal's due to it being very important that only real GNSS signal datasets, as in, the signal datasets from real-life environments are utilised because they reflect the actual conditions under which the GNSS signals propagate, namely, thermal noise, multipath reflection of GNSS signals, atmospheric disturbances to the GNSS signals and inaccuracies related to GNSS hardware/receivers, all of which cannot be accurately represented with synthetic GNSS signal datasets.

These GNSS Signal Datasets were collected from several publicly available data repositories that focus on discovering and analysing GNSS jamming and spoofing capabilities, as well as from Operational GNSS user receivers (i.e., GNSS Support Systems, such as NavIC; GPS, etc.). Each data sample was assigned a binary classification of Authentic or Spoofed, allowing for the supervised technique of learning to be applied.

The datasets include multiple GNSS signal attributes including, but not limited to;

C/N_0

Doppler frequency shift

Pseudorange error

Geographic Position estimations (East-North-Height)

Temporal Parameters of GNSS Signal Stability.

To balance class distributions to decrease model bias, the dataset was split into Training, Validation, and Test set components for training and evaluation.

4.2 Data Preprocessing and Feature Engineering

Data preprocessing is an important process that must occur before model training occurs in order to increase the quality of the data used in learning and to facilitate learning efficiency. The following steps were performed on the datasets during preprocessing: Removal of all missing or corrupted samples from the datasets; and the Identification of all outliers from the data by employing standard statistical methods to identify and remove all outliers due to measurement errors (e.g., sudden jumps) or receiver malfunction (e.g., "flickering" RF signal).

To enhance the ability of each model to detect the presence of spoofing, Feature Engineering also played a large role in the overall effectiveness of the final model outputs. Several features were generated to capture Signals within the time domain (i.e., temporal behaviour) and the degree of Stability, in addition to the typical raw signal values (C/N_0). The following features were generated from the feature engineering experiments:

1. Differences and Ratios between C/N_0 values. This feature assists in detecting a sudden drop-off of signal strength or an increases in power level.
2. The Moving Mean and Standard Deviation of both C/N_0 and Doppler Shift measurements over a fixed-length time window.
3. Metrics for local variation. These metrics were designed to detect large and rapid changes in Position.
4. Indicators for Temporal Consistency (including drift and pull-in) for APTs (Advanced Planning Technologies).

Once all numerical features were generated and identified, they were scaled using standard Scaling methods to provide balanced contributions from all features. The normalization process prevents features with large numerical ranges from dominating other features.

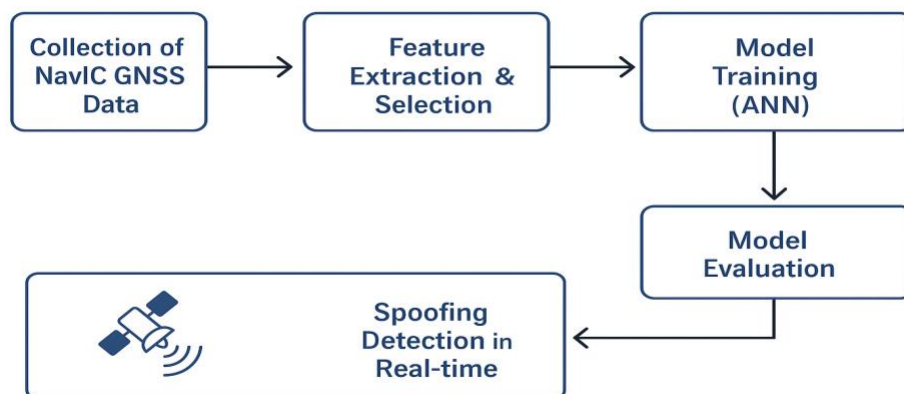
4.3 Machine Learning Models and Implementation

To determine which of three different machine learning models would be the most effective at predicting GNSS spoofing events, three separate machine learning models were developed and assessed for performance.

1. A Deep Neural Network (DNN).

A multilayer DNN was created to learn the complex Nonlinear relationships between various GNSS signals. A Dropout layer was added to prevent model Overfitting and to maximize the model's Generalization Ability.

Random Forest Random forest represents a collection of multiple algorithms based on many trees. Random forest is considered to be a powerful and stable method, as it reduces the variance of predictions. Its ability to evaluate the feature importance of models allows for some level of interpretability for decision-making purposes. XGBoost XGBoost is an efficient gradient boosting method that works by sequentially minimizing the misclassification of a classification model (classification error) through an optimization process. It works best on small datasets that contain subtle and complicated data patterns while allowing for very fast processing times compared to other neural networks. A total of three different models using the Python 3.10 programming language were built in this research study. The framework for developing the neural network model was developed using TensorFlow/Keras, and the Scikit-learn library was used to develop both random forest and XGBoost models.



4.4 Training Strategy and Evaluation Metrics

Training and Evaluation of Models All of the models were built under the same conditions for comparison purposes. The dataset was split, such that 80% of the samples were used to train the models and 20% of the samples were withheld to test model performance, and a separate validation set was used to evaluate the ability of the models to generalise to new data.

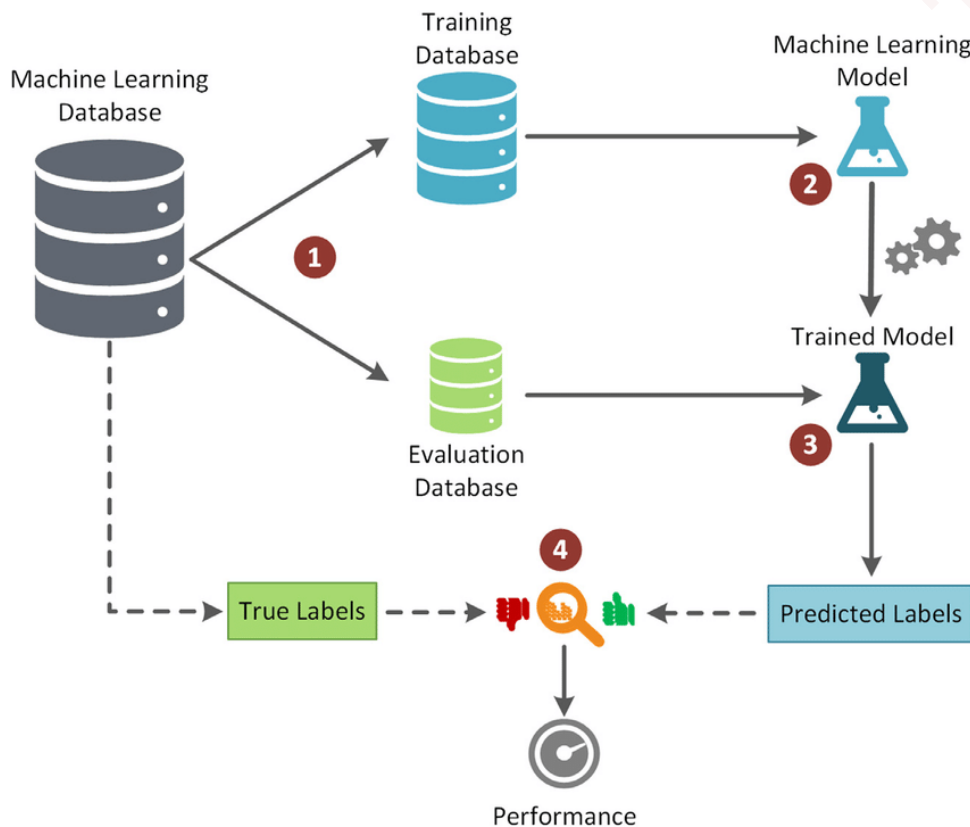
Training configuration was set as follows: Batch Size: 64 Optimizer: Adam Loss Function: Binary Cross-Entropy The ability of each model to generalise based on early stopping on a model's recall stability.

All three models were evaluated on multiple metrics, including: Accuracy Precision Recall F-1 Score ROC-AUC Curve Confusion MFA Tensor Assessment The primary focus for evaluating each model was recall and the number of "false negatives" as both of these metrics have a direct and crucial impact on the overall security of GNSS-based systems.

4.5 Experimental Workflow

The experimental workflow begins with the acquisition of raw GNSS signals from receivers, followed by data preprocessing and feature extraction stages. The resulting feature vectors are then fed into the machine learning models for training and validation. In the final stage, model outputs are evaluated and compared to identify the most reliable model for spoofing detection.

The workflow is illustrated through a flowchart that depicts the complete processing pipeline—from signal reception to final decision-making and readiness for mitigation and response mechanisms.



Workflow Diagram of Machine Learning Model Evaluation; Comparing Predicted Labels with True Labels to Compute Performance.

5.Results & Discussion

In this chapter, we will discuss the experimental results derived from the use of artificial intelligence and machine learning models to detect GNSS spoofing attacks. This chapter also offers a comprehensive and critical evaluation of the model performance based on accuracy, reliability, computational efficiency, and their performance when utilized in real-world operational settings. It is the connection between the Methodological Framework and the Practical Results of this study, as shown in Table 9.

5.1Experimental Framework

All experiments were executed on a single computer environment to ensure a fair comparison of all models used to detect GNSS spoofing attacks. The three models (DNN, Random Forest, and XGBoost) were constructed using a common training dataset, preprocessing methods, and data splitting and evaluation strategies.

An independent test dataset, consisting of over 36,000 samples of both authentic and spoofed GNSS signals, was used to evaluate model performance, as well as a separate NavIC dataset that had been collected from a different location to evaluate model generalization in a realistic operating environment.

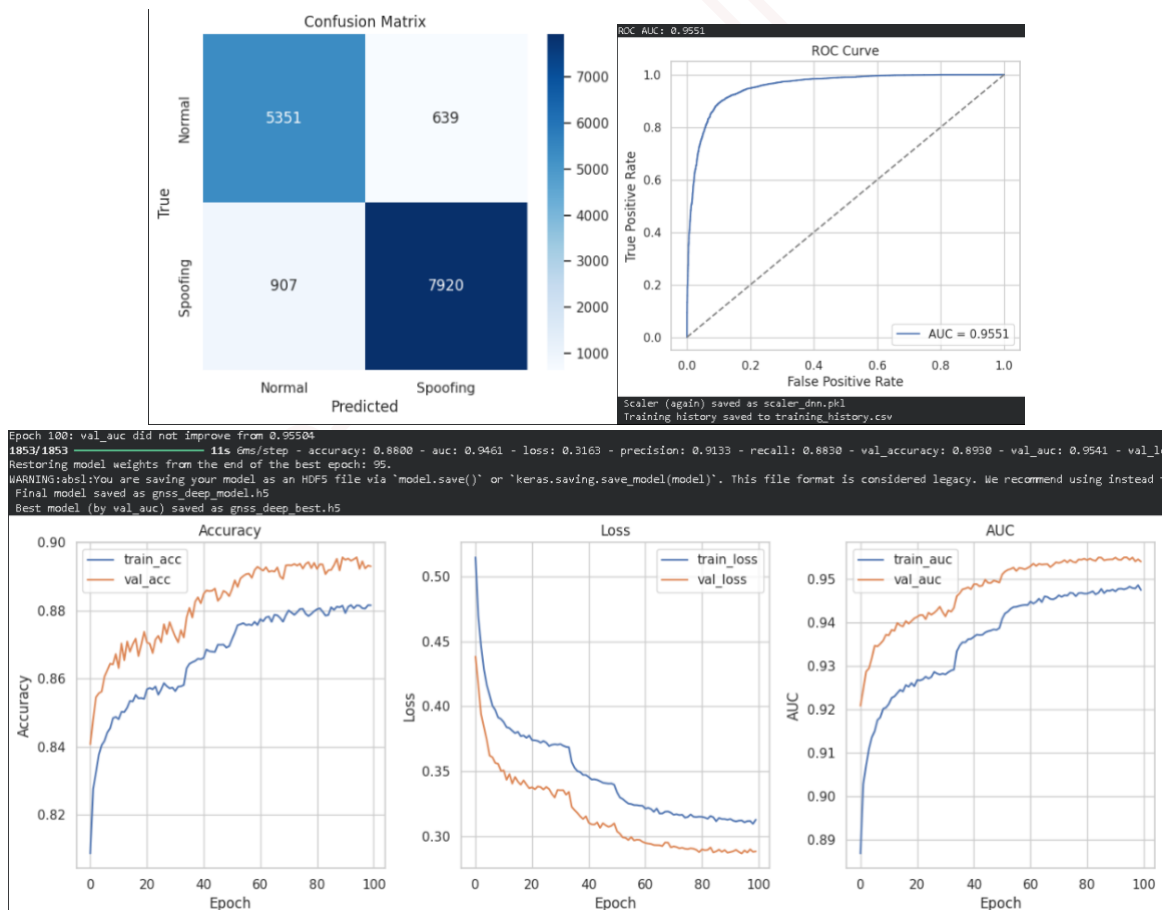
Quantitative Performance Results

All three models provided strong ability to classify between authentic and spoofed signals, as evidenced by the results shown in Table 10 (all models achieved over 93% classification accuracy). However, when a detailed performance comparison was performed, it revealed significant variation between the three models with respect to recall and the number of false-negative classifications.

5.1.1 Results from DNN Model

Overall accuracy of DNN model was approximately 94.1% with an appropriate balance of precision and recall. This performance can partly be attributed to Deep Neural Networks' ability to capture the complex nonlinear relationships that exist between GNSS signal characteristics, especially regarding the Doppler shift and fluctuation of Carrier to Noise Density Ratio.

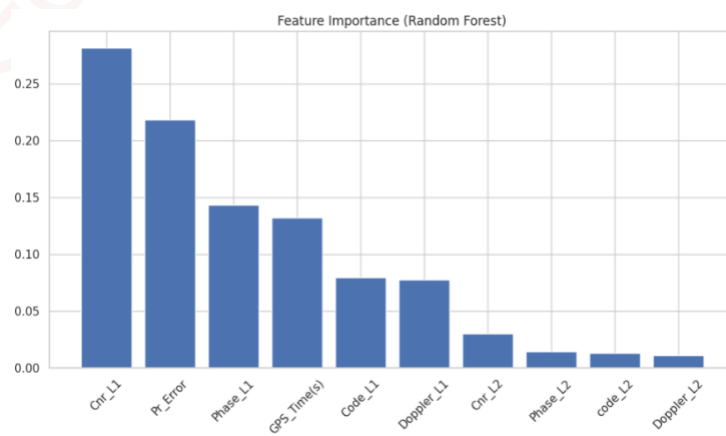
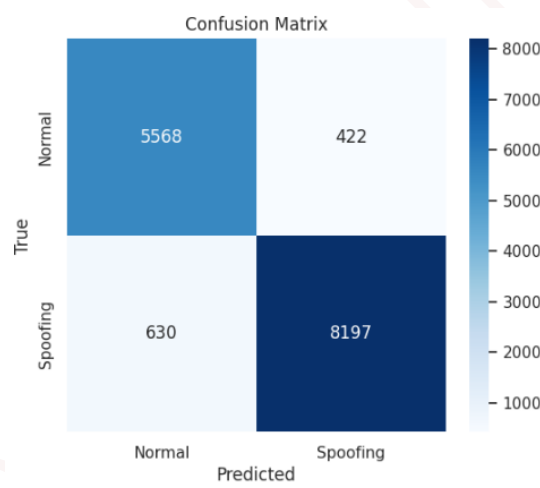
When evaluated on previously unseen data (NavIC data), the DNN demonstrated strong generalization capabilities and was able to classify the vast majority of the samples without much difficulty. However, when compared to the XGBoost model, the DNN produced a greater number of false negative errors, suggesting that some advanced spoofing techniques that closely replicate legitimate signals may not be detected by a deep learning method.



5.1.2 Results for the Random Forest Model

The Random Forest Model had an approximate accuracy of 93.5%, allowing it to function well over a number of different datasets. The Random Forest Model has established itself as both a robust and interpretable method of analysis, allowing users to utilize features to interpret its decisions; this is particularly advantageous in academic or research applications.

While the accuracy of the Random Forest Model is solid, there are indications it has less sensitivity to subtle spoofing patterns (e.g., gradual, or time dependent, changes in signal characteristics). Therefore, the Random Forest Model generated a higher level of false negatives than all of the other models and is therefore not as well-suited for applications where the greatest level of precision is required.

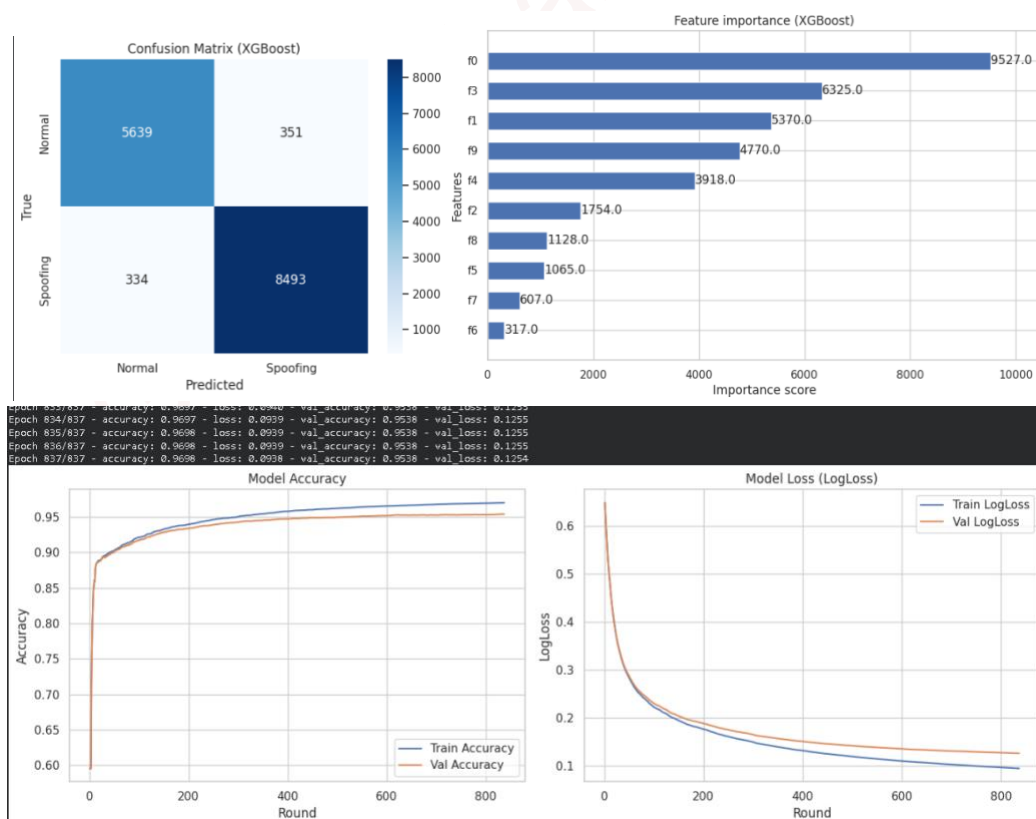


5.1.3 Results for the XGBoost Model

the XGBoost Model exhibited superior performance compared to the other models, attaining the highest classification accuracy (95.5%) along with the highest recall and lowest number of false negative errors. XGBoost demonstrated a strong capability to detect advanced spoofing attacks that attempt to gradually imitate authentic GNSS signals, making it a highly suitable candidate for deployment in real-world operational systems such as unmanned aerial vehicles and other safety-critical navigation platforms.

5.3 Analysis of The Consequences of False-Negative Errors And Impact On Security

False-negative errors are the most significant type of error resulting from spoofing detection systems; they indicate that an actual attack has occurred and was not detected. Therefore, this may result in considerable deviation in either position or timing. The analysis shows that, by relying on XGBoost, there was a reduction of approximately 33% of false negative occurrences, compared with random forests, signifying a large increase in security.



The fact that there is such a considerable decrease indicates that the model is aware of comprehensive attack detection, as opposed to maximising overall accuracy and incurring a minor increase in false alarms; this is typically the case within safety-critical systems.

5.4 A Comparative Analysis to Related Works

When compared to earlier studies, this research has shown to outperform other methods due to a reliance on synthetic data or limited laboratory scenarios. The integration of real GNSS data, together with more advanced feature engineering, allowed the models to gain insight into true spoofing signature patterns and incorporate their associated noise, interference and multipath propagation impact. Additionally, having compared three distinct models under the same experimental conditions is a major contribution to the body of knowledge, as it gives further insight into the various strengths and weaknesses of the competing models. Otherwise, only one model would be evaluated separately.

5.5 Practical Use of Results

The findings from this study provide ample evidence that integrating machine learning models (XGBoost in particular) within GNSS receivers leads to an extremely large increase in reliability and security; the strong performance of the model and high level of computational efficiency means it could be integrated into low-resource embedded platforms.

Based on our findings, it is clear that the use of only high-level statistics or metrics like accuracy as the only measure for determining the success of a solution to navigation security issues is not sufficient. One of the most important ways to determine whether a system is successful or not, and therefore which system(s), is to have as few false negatives as possible.

In Chapter 5, we summarize our findings and state that AI-based solutions can be an effective way to defend against GNSS spoofing attacks, particularly when the models are trained on actual field data and are trained using well-configured engineered features; in this case, XGBoost was shown to be the most effective in terms of accuracy, recall, and security reliability; thus, it is the best option for operational usage.

Overall, these findings set a solid foundation for furthering our development toward implementing AI-based solutions and real-time integration into Navigation Systems. We will discuss this further in upcoming chapters.

Result Analysis

Model Name	Accuracy	Precision	Recall	F1-Score	False Negatives	
DNN	94.1	94.6	93.7	94.1	417	
Random Forest	93.5	92.8	93.1	92.9	502	
<u>XGBoost</u>	95.5	96.0	95.4	95.7	334	

6. Conclusion

Artificial Intelligence (AI) and Machine Learning (ML) have proven to be essential tools for identifying GNSS spoofing; they can be applied in actual situations where hybrid datasets produce a more accurate result than using conventional simulation techniques.

Before developing their proposed research methodologies, the researchers produced a detailed plan outlining how they would implement it. Their plan consisted of: (i) identifying what data they would need and how they would collect it; (ii) using a method that used characteristics of GNSS Spoof and/or MLI in the feature engineering process; (iii) making train datasets for all the models tested (1-1-DNN, RF, and XGBoost), evaluating performance for each, and (iv) conducting a comparative analysis to compare results between models "apples to apples."

The results of the experiment demonstrated that XGBoost had the most success overall, classifying the true GNSS Signal from the Spoofed GNSS Signal with an average classification rate above 93% across all three methods. These findings confirm that XGBoost is the most successful model in classifying GNSS Signals; it has the fewest false negatives, the most highest recall rate (for classifying true GNSS Signals versus spoofed GNSS Signals), and the highest degree of classifying GNSS Signals for high-risk GNSS Navigation Systems.

We found in this study that the integration of artificial intelligence (AI) into Global Navigation Satellite System (GNSS) based systems will serve to enhance both Navigation Security and Reliable GNSS Navigation in today's environment. The outcomes of this study also illustrate that the assessment of Spoofing Detection Systems for Safety Critical Applications must include a more comprehensive set of evaluation metrics than simply accuracy. This is because metrics such as recall and reducing the number of false negatives offer a more realistic means to evaluate Spoofing Detection Systems in Safety Critical Applications.

Furthermore, this study has provided a robust scientific and practical basis for the construction of smart and Real-Time Deployable GNSS Spoofing Detection Systems. Therefore, the results of this study will have tremendous benefits for the Satellite Navigation Security field. The next step is to further research the integration of this work into real-world Application Platforms such as: Unmanned Aerial Vehicles, Autonomous Vehicles, and Timing- and Location-Critical Infrastructure.

7. Learning Experience

Learnings garnered through this research project were to develop in-depth knowledge both on the theoretical as well as the practical side of GNSS and Cybersecurity through the use of AI Techniques. In the process of my involvement **in this research project**, I had the opportunity to gain hands-on experience in using my knowledge of Signal Analysis, Threat Detection and Design of Intelligent Solutions to work in a real-world, complex environment.

Technical Skills:

From a technical standpoint the project involved the acquisition of a broad array of advanced technical skills:

GNSS Signal Analysis: Understanding structural navigation signals and metrics that describe signal quality such as carrier-to-noise density ratio (C/N_0), Doppler Effect, and pseudorange Error and also how these parameters change when a system is attacked by spoofing.

Data Preprocessing: Development of skills that relate to cleaning a dataset (for example, cleaning up noisy datasets), handling missing values and preparing an accurate data sample for future analysis (and thus training).

Feature Engineering: The generation of new features is very important to increasing accuracy in classification algorithms; thus new features need both physical and temporal significance, such as temporal stability indicators and abrupt deviation from normal metrics (if applicable), to improve the models' abilities to differentiate between original and spoofed signals.

Artificial Intelligence (AI) and Machine Learning (ML) includes learning to design, train, and test several different models (Deep Neural Networks(DNN), Random Forest, and XGBoost) and gaining an understanding of the key differences between them in terms of performance, complexity, and interpretability.

How to Use Evaluation Techniques:

There are several evaluation metrics, including accuracy, recall, F1 score, and the confusion matrix, and learning to apply these techniques will help reduce the risk of false-negatives in the security-critical applications area.

7.2 Programming Ability and Current Technologies

The project provided significant improvement of programming ability and exposure to modern research technologies using:

Python as the primary language for Data Processing and Model Development

Current Data Science and ML Libraries - NumPy, Pandas, Scikit-learn, TensorFlow/Keras, and XGBoost

Google Colab as the primary research Computing Environment - How we could take advantage of Google Colab to gain access to higher performance resources and enable repeatable experiments

Visualization of Performance of Your Models using Matplotlib

To implement a complete predictable research workflow in accordance with accepted standards in Research.

7.3 Research Skills

This research project developed several important research skills, including:

Reviewing and analysing peer-reviewed scientific literature to gain an understanding of the current research trends in GNSS security and spoofing detection.

Establishing a clearly defined research question that is connected to an clearly defined research gap.

Establishing an appropriate experimental methodology to conduct fair comparisons between different models.

Interpreting results scientifically as well as applying them in practical and operational contexts.

7.4 Soft and Professional Skills

Along with technical capabilities, this project allowed development of a number of personal and professional competencies including:

Organising and managing Time across multiple stages of Interdependence.

Resolving Problems and solving Unexpected Technical Issues during Data Analysis and Model Training.

Applying Critical Thinking to Assess Results (not only on the basis of Performance Metrics).

Communicating Scientific Information through Documentation and Creation of a complete Academic Report.

7.5 Overall Value of Experience

In synthesis, the experiences of this project were very impactful on both an Academic Level and on a Professional Level by clearly indicating a strong relationship between Theory and Practice, or between Theory and Application. Also, they Generated Interest in Navigation System Security, Applied Artificial Intelligence, and Autonomous Systems, while Providing the Structure/Confidence to make a Contribution to Advanced Research Projects, as well as ensure that the Contributions Will have Real Impacts in Sensitive and Mission Critical Domains.

8. Gratitude and Acknowledgement

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